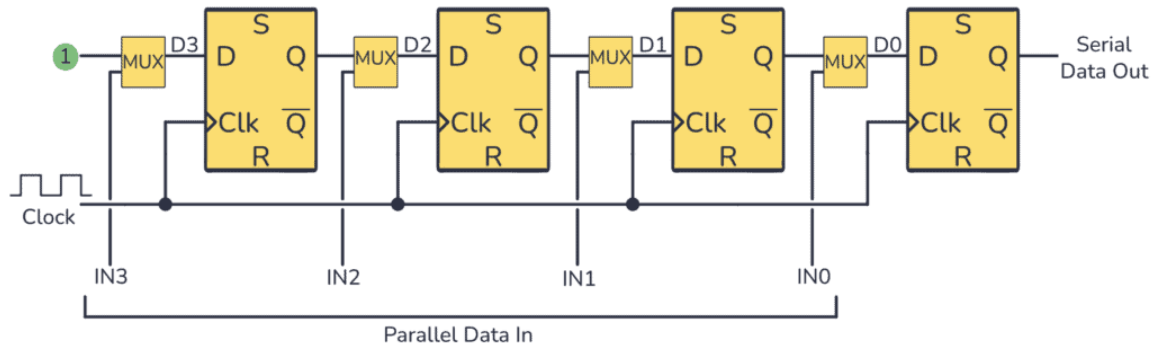


Day 33

4-bit Parallel-In Serial-Out (PISO) Shift Register in Verilog

Circuit Diagram :



Verilog Code

verilog code:

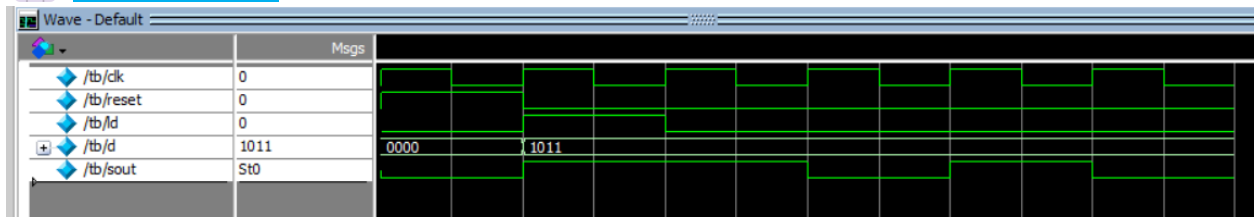
```
1 module PISO(clk,reset,ld,d,sout);
2 input [3:0]d;
3 input clk,reset,ld;
4 output sout;
5 reg [3:0]Q;
6 always @(posedge clk)
7 begin
8 if(reset) Q <= 4'b0000;
9 else if(ld) Q<=d;
0 else
1 Q<=(Q>>1);
2 end
3 assign sout=Q[0];
4 endmodule
```

Testbench Code

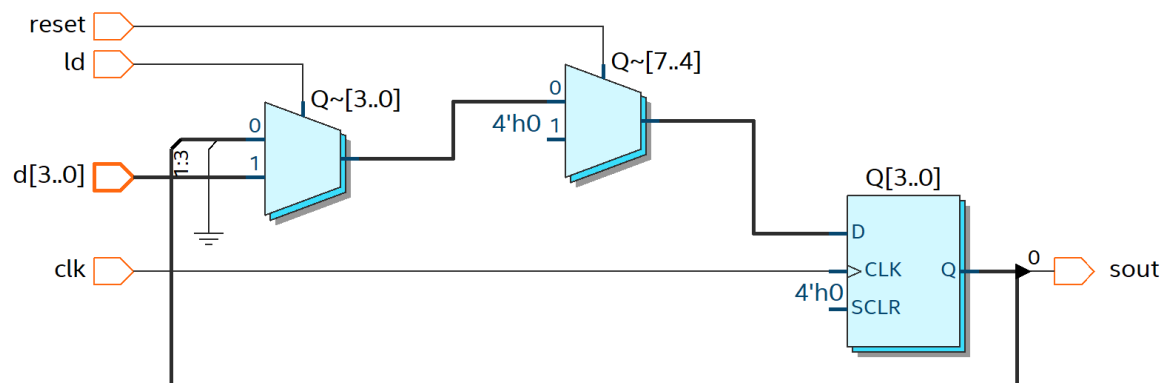
Tb code :

```
1  module tb;
2  reg clk,reset,ld;
3  reg [3:0]d;
4  wire sout;
5  PISO dut(clk,reset,ld,d,sout);
6  initial begin
7  clk=0;
8  forever #5 clk=~clk;
9  end
10 initial begin
11 reset=0; ld = 0; d = 4'b0000;#5
12 reset=1; #10
13 reset=0; d=4'b1011; ld=1; #10;
14 ld = 0;
15 #40
16 $finish;
17 end
18 endmodule
```

Waveform



Schematic





EDA Tools Used

- IntelQuartusPrime
- ModelSim

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